

$$\text{Trade Rate } \alpha_t = \frac{Y}{V_t} \quad (8.7)$$

$$POV_t \quad \alpha_t = \frac{Y}{Y + V_t} \quad (8.8)$$

Important Note: for our analysis we are using the market impact formulation with the trade rate strategy with $a_4 = 1$. This is equivalent to the single period trade schedule formulation discussed in Chapter 7, Advanced Algorithmic Forecasting Techniques.

ALGORITHMIC DECISION MAKING FRAMEWORK

The algorithmic decision making framework is about traders instructing the algorithm to behave in a manner consistent with the investment objectives of the fund. If traders enter orders into an algorithm without any pre-specified rules, or with rules that are not consistent with their investment objective, the only thing we can be certain of is that the algorithm will not achieve best execution. Of course the algorithm may realize favorable prices at times, but this would be due to luck rather than actual intentions. Best execution is only achieved through proper planning.

Best execution is evaluated based on the information set at the time of the trading decision (e.g., ex-ante). Anything else is akin to playing Monday morning quarterback.

The algorithmic decision making framework consists of:

1. Select Benchmark Price
2. Specify Trading Goal (Best Execution Strategy)
3. Specify Adaptation Tactic

1) Select Benchmark Price

Investors need to first select their benchmark price. This could be the current price which is also known as the arrival price, a historical price such as the previous day's closing price, or a future price such as the closing price on the trade day. These are described as follows:

Arrival Price Benchmark

The arrival price benchmark is often selected by fundamental managers. These are managers who determine what to buy and what to sell based on company balance sheets and long-term growth expectations. These managers may also use a combination of quantitative and fundamental